

Data-Driven Turbulence Closure Using a Tensor Basis Neural Network without Reliance on Baseline Models

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Introduction

We introduce a novel machine-learning-based turbulence closure approach in which a Tensor Basis Neural Network (TBNN) is embedded directly into a Reynolds-averaged Navier–Stokes (RANS) solver without reliance on any traditional baseline turbulence model. The TBNN is trained to predict the Reynolds stress anisotropy tensor from local invariant and geometry-informed features. Its output is subsequently processed by a regression model that computes an optimized eddy viscosity field, which is then coupled into the RANS momentum equations as a zero-equation turbulence closure. The model is evaluated on three turbulent flows over complex geometries — a four-period wavy-bottom channel, a curved step, and a rectangular step. The inclusion of geometry-informed features enhances model robustness, yielding numerically stable and converged solutions across all cases. The predicted velocity fields and turbulence distributions closely align with large-eddy simulation data, confirming the accuracy of the proposed framework and demonstrating its capability to operate independently of conventional turbulence closures.

Methodology

RANS equation:

$$\frac{\partial U_i^*}{\partial t^*} + U_k^* \frac{\partial U_i^*}{\partial x_k^*} = \frac{-1}{\rho} \frac{\partial P^*}{\partial x_i^*} + \frac{1}{Re} \frac{\partial^2 U_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial}{\partial x_j^*} \tau_{ij}^*$$
$$u_i^* = U_i^* + \langle u_i^* \rangle \quad \tau_{ij}^* = \langle u_i^* u_j^* \rangle \quad * : \text{Dimensionless}$$

Tensor basis representation, (Pope 1975):

$$b_{ij} = B_{ij}(S_{ij}^*, R_{ij}^*, \psi, \phi)$$
$$\Rightarrow \sum_{k=1}^{10} G^{(k)}(\lambda_n | \mathbf{n} = \{1 \sim 5\}, \psi, \phi) T_{ij}^{*(k)}$$

Optimal eddy viscosity:

$$b_{ij} = \sum_{k=1}^{10} G^{(k)}(\lambda_n | \mathbf{n} = \{1 \sim 5\}, \psi, \phi) T_{ij}^{*(k)} \approx -2\nu_t^* S_{ij}^*$$

$$\nu_{t,opt}^\dagger = \arg \min_{\nu_t^* \geq 0} \|b_{ij} - (-2\nu_t^* S_{ij}^*)\|$$

$$\tau_{ij}^* = b_{ij} + \frac{1}{3} \delta_{ij} = b_{ij} \Rightarrow -2\nu_{t,opt}^\dagger S_{ij}^*$$

Motivation & Objectives

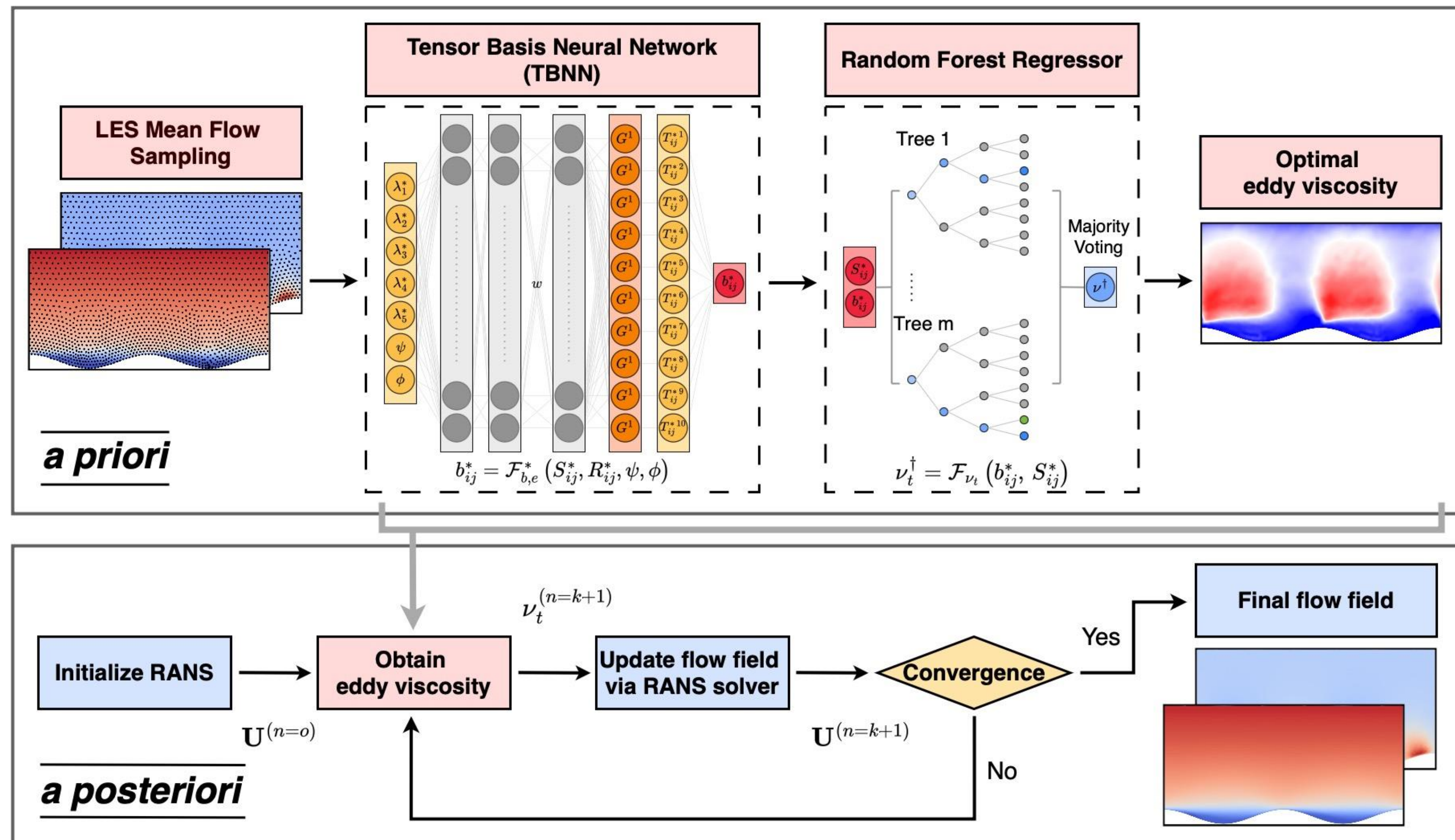
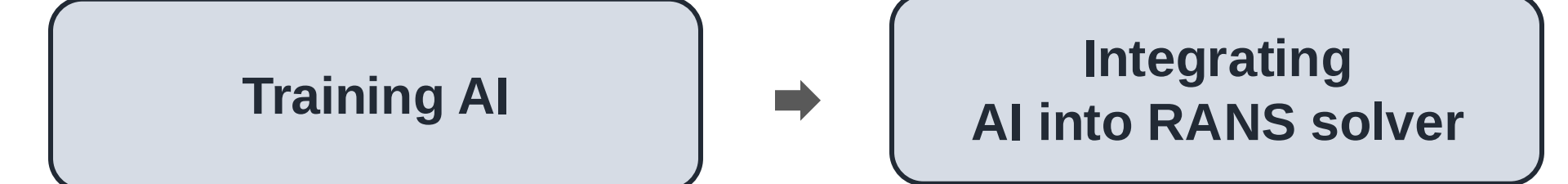
Past methods for tensor basis representation



- Computationally inefficient
- Ignores the inherent fitting capability of AI

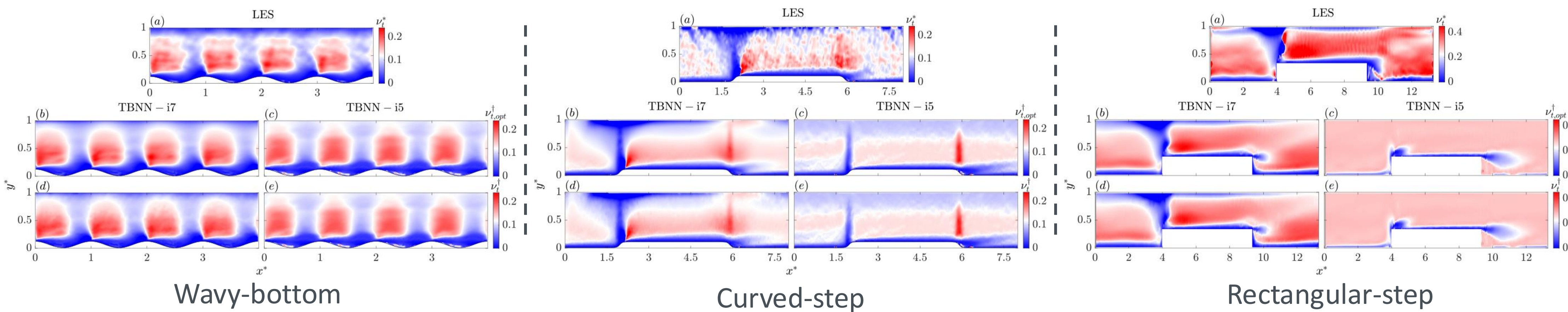
Our method

- Using AI to predict anisotropic Reynolds stresses
- Integrating AI model into RANS solver



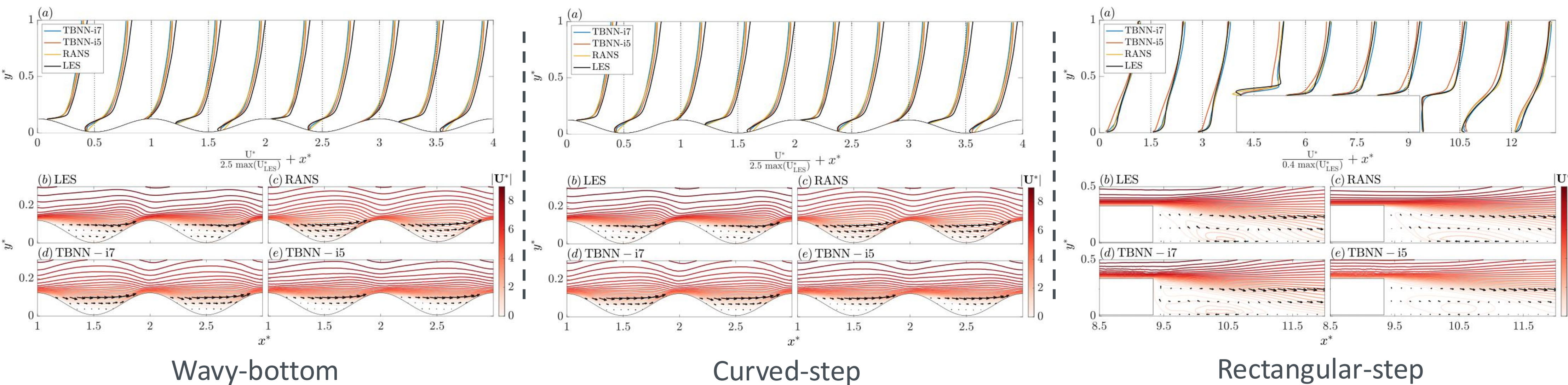
Results & Discussion

(i) Effect of Geometry-Informed Features on Eddy Viscosity Prediction (*a priori*)



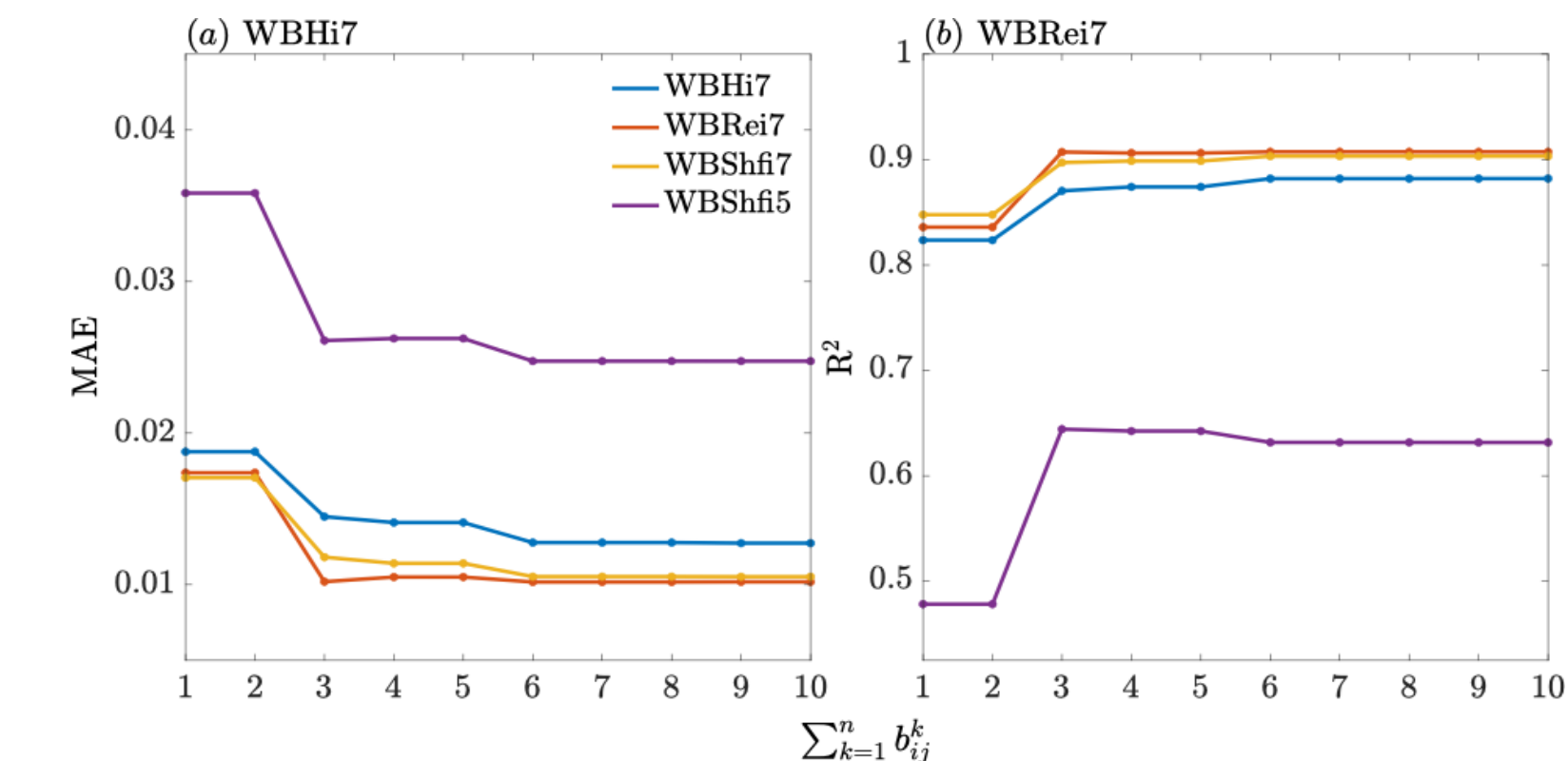
- Adding geometry-informed features improves eddy viscosity prediction, especially near boundaries where the original tensor basis representation cannot fully capture geometric effects. This improvement is consistently observed across all three geometries.

(ii) Velocity Field Computation Across Different Approaches (*a posteriori*)



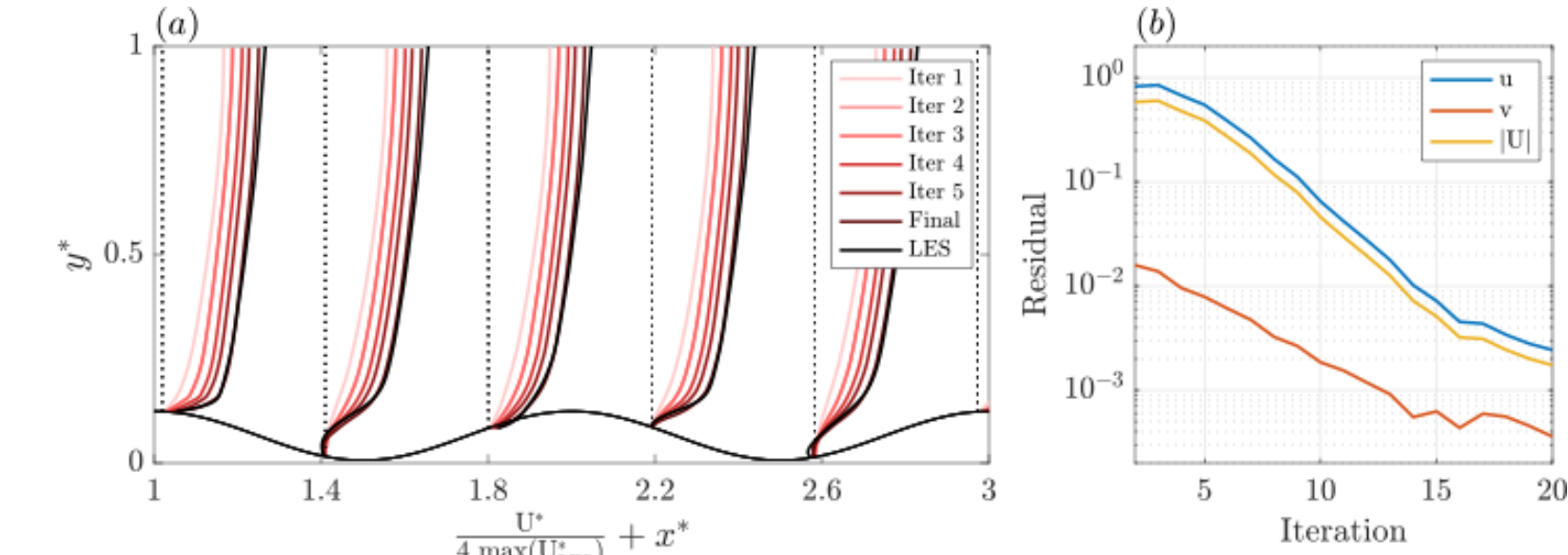
- The model accurately predicts velocity fields across the domain, including near-wall separation zones. Results are consistent with LES and show improved accuracy over baseline RANS models.

(iii) Prediction Accuracy vs. Tensor Order

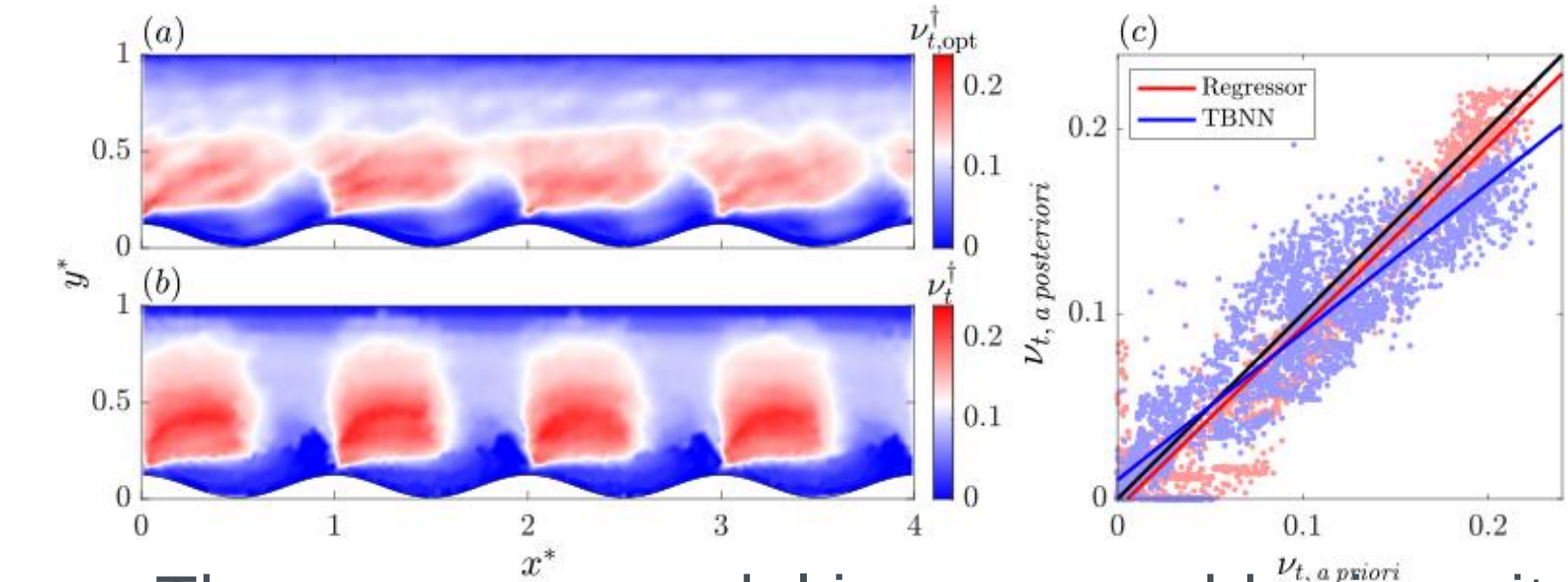


- Model accuracy improves with tensor order and becomes stable by third-order.

(iv) Flow Field Development and Convergence



(v) Comparison of Eddy Viscosity Evolution Between Models



- The regressor model improves eddy viscosity prediction with more stable iterative behavior.

Conclusion

- This work presents the first baseline-free turbulence model that embeds a TBNN into the RANS solver to compute flow fields.
- Incorporating geometry-informed features improves model robustness and enables better generalization across varying flow configurations.
- The proposed method achieves accurate and stable predictions across three complex geometries: wavy bottom, curved step, and rectangular step.

Reference

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Acknowledgments

This work was supported by the Taiwan National Science and Technology Council (NSTC), Ocean Engineering Division, under grant no. 112-2221-E-002-186-MY3. Computational resources were provided by the National Center for High-performance Computing (NCHC), National Applied Research Laboratories (NARLabs), Taiwan. The source code is available at: <https://github.com/YaoHung0315/Tensor-Basis-Neural-Network>

